Determination of Spring Constant Using Static and Dynamic Methods

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25.12.2024

Objective

To calculate the spring constant k of a vertically hanging spring using two different methods: **static** and **dynamic**, and compare the results with error analysis.

Part 1: Static Method

Setup

- Use a spring vertically suspended from a fixed support.
- Attach a mass of $M = \text{xxx}$ grams to the free end of the spring.

Procedure

1. Record the initial position of the spring without any mass attached by taking a photo.
2. Attach the mass M to the spring and allow it to settle into equilibrium.
3. Record the final position of the spring by taking another photo.

Measurement

- Use the ruler provided on the setup to measure the displacement of the spring Δy from the photos, see Figure 1.

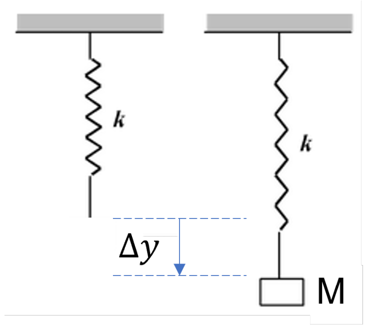

Figure 1: Measurement of the elongation of the spring in the static setup.

- Calculate the weight W of the mass:

$$W = Mg$$

where $g = 9.81\,\mathrm{m/s}^2$ is the gravitational acceleration.

- Determine the spring constant k using the formula:

$$k = \frac{W}{\Delta y}$$

Uncertainty Calculation

By driving chatGPT with the following prompt or your own modified prompt, determine the mean expected stiffness and it's uncertainty

```
Propose a method to evaluate the stiffness of a spring
by using photos of its initial and deflected states under
a static load, with no more than 10 samples from each state,
with results including histograms, the mean and standard
deviations of deflections, the expected stiffness value
and it's uncertainty.
```

Record chatGPT's response, and report it along with the prompt for future reference. If not sure about the meaning of an expected value and it's uncertainty, please ask chatGPT to define them for you.

Part 2: Dynamic Method

Setup

- Use the same spring-mass system.
- Attach a different mass $m = \mathrm{xxx}$ grams to the spring.

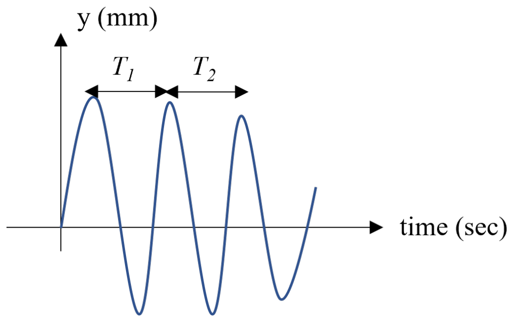

Figure 2: Vertical oscillation of the spring-mass system recorded in the dynamic setup

Procedure

1. Pull the mass down slightly to stretch the spring and then release it to initiate vertical oscillation.
2. Record the motion of the oscillating mass using a video camera.

Video Analysis

- Use the **Tracker Video Analysis Tool**: `https://physlets.org/tracker/`.
- Analyze the video to obtain the position-time graph of the mass's vertical motion.

Calculation

- Identify the period of the oscillation for the first two complete cycles T_1 and T_2, see Figure 2.
- Use the formula to calculate the spring constant for each period:

$$k = \frac{4\pi^2 m}{T^2}$$

- Calculate the average k_{dynamic} from the two values obtained.

Comparison and Error Analysis

Compare the static spring constant k_{static} with the dynamic spring constant k_{dynamic}. Use the following formula to calculate the relative error E:

$$E = \left| \frac{k_{\text{dynamic}} - k_{\text{static}}}{k_{\text{static}}} \right| \times 100$$

Interpret the results and discuss potential sources of error (e.g., measurement inaccuracies, damping effects, or alignment issues).